

DBMS Assignment 11

Instructions:

- Submit one **.ipynb file**
- The pdf file should contain the question, query/code, outputs, and explanation.
- The name should be **<roll no>_Assignment11.ipynb**
- Complete the exercise before 4.45 PM and submit the files in Moodle.
- Marks Distribution for both Postgres and Python **(query/code + output+explanation)**

Total Marks = 30

17/04/2026

Note: All questions refer to a database called company_db and a collection called employees containing the following documents:

You can extract the records from the .json file using:

```
with open("file_path") as f:
    records = json.load(f)
```

1. (a) Import PyMongo and create a MongoClient object. Connect it to a database called company_db and create a collection called employees. (1)
(b) Insert all employee documents from .json file into the employees collection using a single PyMongo command. (1)
2. Write a PyMongo command to retrieve and print a few documents(limit 5) from the employees collection.(Do not print all documents this might crash your notebook) (1)
3. Write commands to:
(a) count the total number of documents in the collection, and (1)
(b) count the number of employees whose dept is "HR". (1)
4. (a) Update Nisha's(Nisha Patel) salary to 50000 using update_one(). After updating, print the matched_count and modified_count. (1)
(b) Delete the document where the name is "Kiran Swamy" using delete_one(). Print the number of deleted documents. (1)
5. (a) Retrieve the name and salary of all employees whose salary is greater than 1,00,000, sorted by salary in descending order. Use sort() chained on the cursor. Do not display _id. (2)
(b) Write a single query using a logical operator to find all employees who are either in the "IT" department or have a salary greater than 1,20,000. Print their name, dept, and salary. (2)

6. Give every employee in the "HR" department a salary raise of 15,000 using `update_many()`. Then use `$push` to add the skill "Leadership" to all HR employees.(2)
7. Write a query to find all employees who have at least one skill that is alphabetically between "Java" and "SQL" (i.e., `>= "Java"` and `<"SQL"`). (3)
8. Write an aggregation pipeline using `$unwind` on the skills array to find the top 3 most common skills across all employees. The output should display skill (the skill name) and count (how many employees have it), sorted by count descending. (4)
9. Write a query to find all employees who are neither in the "Finance" department nor located in "Mumbai" (check inside nested address) nor have a salary above 100000. Return their name, dept, address.city, and salary. (3)
10. Write a query to find all employees whose name starts with the letter 'S' and whose city (inside address) ends with the letter 'i' (e.g., Delhi, Mumbai, Kochi). Return name and address.city only (3)
11. Write an aggregation pipeline to compute, per department, the following statistics for employees who joined after 2015. (4)
 - Total number of employees
 - Average salary (rounded to 2 decimal places using `$round`)
 - Maximum rating
 - Number of managers (sum of `is_manager` where true counts as 1)